

Experiment 1:

Aim: To demonstrate confusion matrix using python

Program:

```
import numpy as np
from sklearn.metrics import confusion_matrix
import matplotlib.pyplot as plt
import seaborn as sns

y_true = np.array([0, 1, 0, 1, 0, 1, 0, 1, 0, 0, 1, 1, 1, 0, 0])
# Predicted labels
y_pred = np.array([0, 1, 1, 0, 0, 1, 0, 1, 1, 0, 1, 1, 0, 0, 0])

cm = confusion_matrix(y_true, y_pred)

print("Confusion Matrix:")
print(cm)

plt.figure(figsize=(6, 5))
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues", cbar=False,
            xticklabels=['Class 0', 'Class 1'],
            yticklabels=['Class 0', 'Class 1'])
plt.xlabel('Predicted Label')
plt.ylabel('True Label')
plt.title('Confusion Matrix Demonstration')
plt.show()
```

